LifeOS Domains: Implementation Plan and App Flow

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Overview

This plan says how to build each of the 21 LifeOS domains and how a user moves through each one in the app. It adds domains one wave at a time on top of the base system: the Next.js app, the Postgres database, the calendar service, the chat orchestrator with its typed tools, and the domain template.

Each domain entry has five parts.

- **Data:** the new tables the domain needs.
- **Rules and tools:** the rule functions in `lib/calendar` or the domain folder, and the agent tools that wrap them.
- **Jobs:** scheduled or background work.
- **App flow:** the numbered path a user takes through the screens and chat.
- **Done check:** one test that proves the domain works.

Assumptions.

- The base system exists: calendar service, `api/chat`, undo, confirmation, cron endpoints protected by `CRON_SECRET`.
- Domain names for tools are prefixed, such as `education_add_exam`, and every tool is a thin wrapper over a tested function.
- Estimates are in weeks, not dates, so you can start whenever the base is ready.

The domain build template

Every domain follows the same eight steps in the same order, so the first four domains prove the pattern and the rest are mostly configuration plus their own rules.

1. **Config.** Add the domain row: name, color, tier (1, 2 or 3), goals, default event types and the prompt file. Put the domain behind a flag, `DOMAIN_<NAME>_ENABLED`.
2. **Tables.** One additive migration with only this domain's tables, each carrying the user and filtered by the logged-in user.
3. **Rule functions.** Write the domain's rules as plain functions with unit tests, with no model involved.

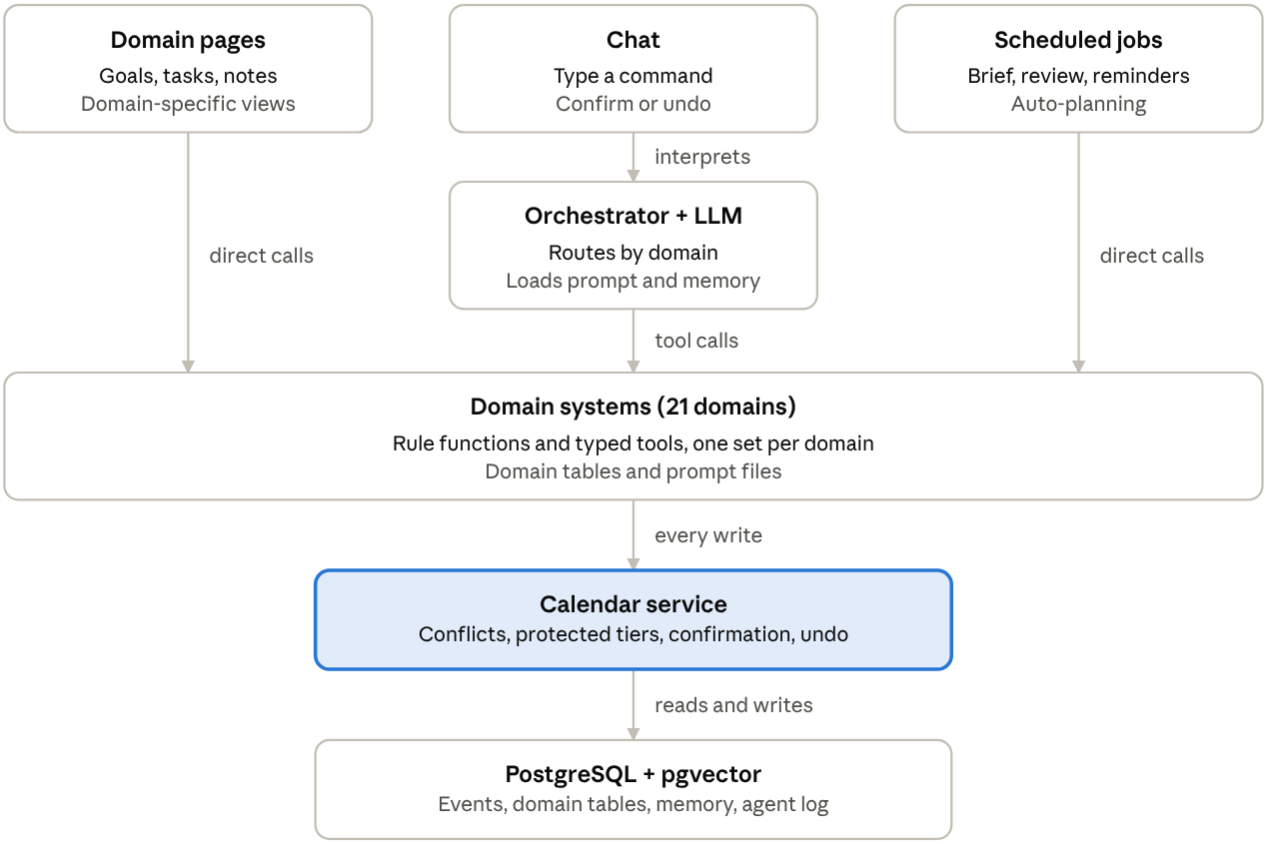
4. **Tools.** Wrap each function as a typed tool with a zod schema, and register it for this domain only.
5. **Prompt.** Write the domain prompt file: vocabulary, what the domain may propose, what it must confirm.
6. **Screens.** Build the domain page: goals, task list, notes, mini calendar, and the domain's own views.
7. **Jobs.** Add any cron or background action, protected by `CRON_SECRET`.
8. **Test and ship.** Run the domain's saved commands, safety cases and undo test, then turn the flag on.

Definition of done for a domain. Its done check passes, every write can be undone, its events appear in the right color in the shared calendar, and with the flag off nothing about it is reachable.

App flow

All 21 domains plug into the app the same way: three entry points, one set of domain systems, and one calendar service that every write must pass.

Every domain screen, chat command and job reaches data through one calendar service



app flow · 3 entry points, 1 calendar service, 1 database

Chat is the only entry point that goes through the model. Domain pages and scheduled jobs call the domain rule functions directly, so the same conflict, tier and undo rules apply however a change starts.

What the user sees, in order

1. Opens a domain from the Domains page (or types in chat) and sees that domain's goals, tasks and mini calendar.
2. Adds or changes something; the domain system proposes events or tasks.
3. A draft appears, as dashed blocks or a preview, with Accept, Edit and Skip.
4. On Accept, the calendar service checks conflicts and tiers, saves, and shows an Undo button.
5. Check-ins and results feed the domain's metrics, the morning brief and the weekly review.

Wave 1: prove the pattern (Education, Competitions, Karate, Sleep)

These four set the shared rules (protected time, backward planning, slot ranking), so they are built first and slowest. Estimate: 3 weeks, with Sleep first because the others depend on its minimum.

Sleep

- **Data:** SleepSetting (bedtime, wake time, minimum hours), SleepLog (date, hours), SleepDebt (computed view).
- **Rules and tools:** `sleep_debt` (sum of 7-day shortfalls), `sleep_guard` (blocks changes under the minimum), tools `sleep_set_window`, `sleep_log`, `sleep_repay_plan`.
- **Jobs:** morning log prompt; nightly debt recompute; wind-down reminder.
- **App flow:** Settings or Sleep page → set window → daily log prompt in the brief → debt strip on Today → "Repay plan" draft → Accept.
- **Done check:** move an event into the sleep window and see the warning; log 3 short nights and receive a repay draft.

Education

- **Data:** Course, Topic (parent, weight, confidence), Exam, Assignment (weight, score), PracticeResult.
- **Rules and tools:** `plan_backwards` with spaced revision, `grade_needed`, tools `education_add_exam`, `education_add_assignment`, `education_rate_topic`.
- **Jobs:** on exam save, create suggested revision blocks; nightly confidence decay.
- **App flow:** Domains → Education → Add exam → paste or enter syllabus topics → agent shows dashed revision blocks → Accept all or Edit → topic ratings update the plan.
- **Done check:** create an exam 10 days away and see revision blocks that avoid sleep, then log a weak topic and see more time on it.

Competitions

- **Data:** Competition (rules, eligibility, submission time), TeamMember, ChecklistItem, RehearsalLog, PostMortem.
- **Rules and tools:** `reverse_sprint` (counts back from submission), `rubric_score`, tools `competition_add`, `competition_checklist`, `competition_score_pitch`.
- **Jobs:** build-freeze and rehearsal reminders; overlap warning between two competitions.

- **App flow:** Chat "Hackathon pitch due Friday 6 PM" → competition saved → reverse-sprint blocks proposed → checklist appears → rehearsal reminder → post-mortem prompt after the result.
- **Done check:** a submission time produces freeze, demo and rehearsal blocks in the right order with no clash.

Karate

- **Data:** TrainingClass (recurring), Syllabus item (belt, status), Grading (date, belt), InjuryNote.
- **Rules and tools:** `belt_path`, `hard_session_limit`, tools `karate_set_grading`, `karate_mark_item`, `karate_log_injury`.
- **Jobs:** weekly drill-block refresh; pre-tournament taper.
- **App flow:** Karate page → set grading date → agent splits the syllabus into weekly drills → check items off after classes → taper appears before a linked tournament.
- **Done check:** a grading date 8 weeks out spreads unsolid items across weeks, and an injury note caps hard sessions.

Wave 2: remaining core domains

Each domain reuses the wave 1 template, so the work is its tables, its rules and its prompt. Estimate: 5 weeks for 8 domains. Build Rest and buffer and Forward planning first, because other domains lean on them.

Rest and buffer

- **Data:** BufferRule, LoadLimit, DeloadWeek.
- **Rules and tools:** `insert_buffers`, `load_check`, `recovery_forecast`; tools `rest_set_limit`, `rest_forecast`.
- **Jobs:** Sunday forecast for the coming week.
- **App flow:** Settings → set daily load limit → any save over the limit shows a warning with blocks to drop → Sunday review shows the forecast and an "Add buffer" draft.
- **Done check:** overload a day and see the warning; a high-risk week produces a buffer proposal.

Forward planning

- **Data:** YearlyDirection, MonthlyGoal (owner domain), DomainIntent, Decision (prediction, review date).

- **Rules and tools:** `balance_gap` (intended minus actual hours), tools `plan_set_intent`, `plan_add_decision`, `plan_weekly_review`.
- **Jobs:** Sunday review feeds the balance wheel; decision review reminder on the saved date.
- **App flow:** Review page → balance wheel → 3 largest gaps → proposed next week → Accept, Edit or Skip.
- **Done check:** log hours in two domains and see correct gaps against stated intent.

Career

- **Data:** Application (stage, dates), Contact (last touch), CvVersion, TargetRole, SkillGap.
- **Rules and tools:** `skill_gap` (frequency of skills in saved job posts), tools `career_add_application`, `career_move_stage`, `career_gap_report`.
- **Jobs:** 7-day follow-up task after applying; contact follow-up list in the brief.
- **App flow:** Career page → add application → move stage → agent creates the next task or prep blocks → gap report suggests learning blocks.
- **Done check:** moving an application to "interview" creates prep blocks before its date.

Volunteering

- **Data:** Organization, Shift, HoursEntry, ImpactNote.
- **Rules and tools:** `hours_total`, `crowding_check`; tools `volunteer_add_shift`, `volunteer_log_hours`.
- **Jobs:** certificate task at a hours target; crowding warning when a shift falls in an exam week.
- **App flow:** Chat "Shift Sat 9 to 1" → event with travel blocks → after the shift, add a note → monthly impact summary.
- **Done check:** a shift in an exam week raises the crowding warning.

Open source

- **Data:** Repo, TrackedIssue, TrackedPr, ContributionSlot.
- **Rules and tools:** `issue_fit`, `stale_pr_check`; tools `oss_add_issue`, `oss_rank_issues`.
- **Jobs:** stale pull request task after 5 days; weekly slot refresh.
- **App flow:** Domain page → add issues by hand (GitHub sync later) → ranked list → reserve slot → status updates.
- **Done check:** three issues rank in the expected order and the top one gets a slot.

Food

- **Data:** MealWindow, MealPlan, GroceryItem, DietProfile.
- **Rules and tools:** `day_load_label`, `assign_meals`, `grocery_sum`; tools `food_plan_week`, `food_grocery_list`.
- **Jobs:** Sunday plan draft; prep-block placement.
- **App flow:** Food page → set meal windows → "Plan my week" → heavy days get zero-cook meals → grocery list.
- **Done check:** a heavy exam day gets a zero-cook meal and a batch-cook block lands on the lightest day.

Blogging

- **Data:** Post (stage), Outline, Idea, ChecklistItem.
- **Rules and tools:** `outline_to_blocks`, `harvest_ideas`; tools `blog_add_idea`, `blog_move_stage`.
- **Jobs:** weekly idea harvest.
- **App flow:** Board → move a card to draft → writing blocks appear in focus hours → checklist must be ticked before publish.
- **Done check:** a draft outline creates blocks sized to its sections.

Learning

- **Data:** Skill, Prerequisite, Resource, PracticeLog.
- **Rules and tools:** `unlocked_nodes`, `skill_score`; tools `learn_add_skill`, `learn_next`.
- **Jobs:** daily learning block chosen each morning.
- **App flow:** Roadmap page → add skills and prerequisites → agent picks the next node → block appears → log practice.
- **Done check:** completing a prerequisite unlocks its child and changes the next pick.

Wave 3: suggested domains

Estimate: 3 weeks for 6 domains. Build Habits and reflection and Portfolio first: they unlock the Reset Plan link and the evidence loop.

Habits and reflection

- **Data:** Habit (source, commitment link), HabitEntry, GraceSetting, Reflection.

- **Rules and tools:** `streak_with_grace`, `stack_suggestion`, `drift_check`; tools `habit_add`, `habit_log`, `habit_overview`.
- **Jobs:** evening check-in reminder; daily drift check.
- **App flow:** Habits page → add habit → one-tap evening check-in → streak and grace shown → drift suggestion offers a lighter option.
- **Done check:** two skips inside the grace number keep the streak; three misses in a row raise a suggestion.

Portfolio

- **Data:** Project, EvidenceEntry (skills, visibility), CaseStudy.
- **Rules and tools:** `skill_evidence_map`, `draft_entry_from_event`; tools `portfolio_add`, `portfolio_kit`.
- **Jobs:** evidence auto-capture runs when a competition result, pull request, shift or post is recorded.
- **App flow:** Result recorded in another domain → draft entry appears → Accept or discard → application kit picks evidence for a role.
- **Done check:** a recorded competition result produces a draft entry with the right skills.

Finance

- **Data:** BudgetCategory, Expense, Funding (deadline, status), SavingsGoal.
- **Rules and tools:** `remaining_budget`, `savings_pace`, `funding_coverage`; tools `finance_add_expense`, `finance_add_funding`.
- **Jobs:** backward-planned blocks per funding deadline; monthly coverage view.
- **App flow:** Chat "Scholarship due Dec 1" → funding saved → essay, documents and review blocks proposed → monthly view marks covered or at risk.
- **Done check:** a funding deadline with too little free time is marked at risk. No advice is ever shown.

Admin

- **Data:** RenewalItem, DocumentIndexEntry (name, expiry), ChecklistTemplate.
- **Rules and tools:** `deadline_ladder`, `paper_trail`; tools `admin_add_item`, `admin_build_checklist`.
- **Jobs:** reminders at 30, 14, 7 and 1 days.
- **App flow:** Chat "Apply for a course" → checklist with lead times → each item scheduled backward → reminders on the ladder.

- **Done check:** the longest-lead document is scheduled first and ladder steps already passed are skipped.

Travel and commute

- **Data:** Route, Trip, PackingList, TravelBlock (linked to its event).
- **Rules and tools:** `door_to_door`, `packing_from_trip`; tools `travel_set_route`, `travel_plan_trip`.
- **Jobs:** travel blocks created on event save and removed with the event.
- **App flow:** Save an event with a location → travel blocks appear before and after → the conflict check re-runs → packing list for trips.
- **Done check:** deleting an event removes its travel blocks; a block that now clashes is flagged.

Social and mentors

- **Data:** Person, ContactRhythm, MentorSession, ActionItem.
- **Rules and tools:** `rhythm_due`, `mentor_prep`; tools `social_add_person`, `social_mentor_prep`.
- **Jobs:** soft reminder when a gap passes the rhythm; prep sheet 2 days before a session.
- **App flow:** People page → set rhythm → reminder → after a mentor session, action items become tasks in their domains.
- **Done check:** the prep sheet includes only the domains approved for that mentor.

Wave 4: sensitive domains (Medicine, Prayer, Mood reflections)

Estimate: 3 weeks. The first week builds the sensitive-domain layer; the three domains cannot ship without it.

Sensitive-domain layer (build first)

- **Opt-in flags.** Each domain is off until you turn it on, with a separate switch for whether its content may be sent to the AI model. When that switch is off, the agent sees only time blocks.
- **Generic notifications.** A notification helper that never includes domain content.
- **Report exclusion.** Weekly, Reset and mentor reports skip these domains unless you include them.
- **Distress handling.** A check on Mood reflections and chat text that pauses planning prompts and responds with care and a pointer to someone you trust or a local support line.

- **Per-domain export and delete.** Deleting also removes the domain's embeddings.

Medicine

- **Data:** Medication (as entered from your clinician), DoseSchedule, Stock, Appointment, SymptomLog.
- **Rules and tools:** `dose_anchor` (tier 1 events), `refill_days`, `visit_brief`; tools `medicine_add_med`, `medicine_log`, `medicine_visit_brief`. No tool changes a dose.
- **Jobs:** dose reminders (repeated once, then logged as missed); refill task at the lead time.
- **App flow:** Medicine page → add medication exactly as prescribed → dose anchors appear on the calendar → one-tap taken or skipped → visit brief before an appointment.
- **Done check:** a block that clashes with a dose is moved or questioned and the dose stays; no screen or reply gives dosing or diagnosis advice.

Prayer

- **Data:** TraditionProfile (method, tracked practices), PracticeLog, SpecialDay.
- **Rules and tools:** `compute_times` (location, date, method), `anchor_blocks`; tools `prayer_set_profile`, `prayer_today`.
- **Jobs:** nightly creation of next-day anchors; recalculation when location changes; special-day notices to Food and Sleep.
- **App flow:** Prayer page → choose tradition and method → times appear as anchors → changes of location update them → fasting periods adjust meals and sleep.
- **Done check:** times recompute for a new city and a clashing block gets a move proposal; if times cannot be computed the system asks.

Mood reflections

- **Data:** MoodEntry (mood, energy, tags, note), Prompt, WeeklyPattern.
- **Rules and tools:** `pattern_factors` (needs enough data points), `gentle_mode`; tools `mood_checkin`, `mood_weekly_view`.
- **Jobs:** evening check-in; Sunday pattern summary; gentle-mode proposal.
- **App flow:** one-tap mood and energy → optional note → weekly view → optional gentle mode proposes a lighter tomorrow → Accept.
- **Done check:** patterns appear only with enough data, a distress phrase pauses prompts, and none of it appears in other reports.

Cross-domain systems

These five systems are built inside the wave that first needs them, not as a separate project.

System	Built in	How it is built	Done check
Protected tiers	Wave 1	Add a `tier` column to domains and a `findConflicts` rule that refuses tier 1 overlaps without confirmation and proposes moves for lower tiers	A tier 3 block is proposed for moving, a tier 1 block never is
Life graph	Wave 2	A `Link` table (from, to, kind) written whenever a task, event, note or goal references another; a `why_here` query follows the links	"Why is this block here" returns the goal it serves
Season profiles	Wave 2	A `Profile` table of rule overrides; switching applies and lists them, and switching off restores defaults	Exam season shortens blocks and turning it off restores them
Evidence loop	Wave 3	An event listener on result records in Competitions, Open source, Volunteering and Blogging that drafts Portfolio, Career and Learning entries	A recorded result yields draft entries, nothing saved until accepted
Energy budget	Wave 4	A daily budget computed from sleep, mood and prior load, and a cost per block type; the agent proposes drops when the total passes it	Over-budget day produces a proposal, and a confirmed change is never blocked

Roadmap

All 21 domains take about 14 weeks after the base system is done. Each wave starts only when the previous gate passes. The estimates assume one builder and shift with exams and competitions.

Wave	Domains	Weeks	Total weeks	Gate to start the next wave
1. Prove the pattern	Sleep, Education, Competitions, Karate	3	3	One real week runs on the app with no manual workarounds
2. Remaining core	Rest and buffer, Forward planning, Career, Volunteering, Open source, Food, Blogging, Learning	5	8	All 12 core domains appear in the calendar and chat
3. Suggested	Habits and reflection, Portfolio, Finance, Admin, Travel and commute, Social and mentors	3	11	Evidence loop works and the Reset Plan link reads habits
4. Sensitive	Medicine, Prayer, Mood reflections	3	14	Sensitive-domain layer passes its safety cases first

The Reset Plan can start once Habits and reflection from wave 3 is live, because its Habit Tracker link depends on it.

Testing and launch checklist

Every domain passes the same checks before its flag is turned on.

Layer	What to test	How
Rule functions	Edge cases: midnight, back-to-back blocks, daylight saving, time zones, empty data	Unit tests with Vitest
Tables and routes	Validation, auth, every row filtered by the logged-in user	Integration tests on a test database
Agent tools	30 to 50 saved commands per wave with expected tool calls	A replay script that checks the resulting calendar
Safety	Tier 1 anchors, bulk changes of 5 or more events, undo, prompt injection in event titles	Cases that must always ask or refuse

Layer	What to test	How
Sensitive domains	No advice, generic notifications, excluded from other reports, distress handling	Dedicated cases run before wave 4 ships
Screens	Drag, resize, layer toggles, domain page and chat on a small phone	Playwright plus manual phone checks

Launch checklist per domain

- [ ] Migration applied and reversible
- [ ] Done check passes
- [ ] Undo tested on each tool
- [ ] Events show in the right color in the shared calendar
- [ ] Flag off leaves nothing reachable
- [ ] Cron endpoints reject requests without `CRON_SECRET`
- [ ] Export and delete work for the domain's data